
LOVELY PROFESSIONAL UNIVERSITY

School of Computer Science & Engineering

PROJECT REPORT

From Clicks to Customers

: E-Commerce Behavior Analysis

A Data Analytics Study using Python and Power BI

Domain: Customer Analytics

Data Source: Government Consumer / Customer Datasets

Level: Intermediate

SUBMITTED TO

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Tools Used: Python | Power BI

Libraries: pandas · numpy · matplotlib · seaborn · scikit-learn

August 2026

1. Project Overview

1.1 Project Details

Topic	E-Commerce Customer Behavior Analysis
Domain	Customer Analytics
Dataset Type	Government consumer / customer datasets
Problem Statement	Analyze data to deliver actionable insights for E-Commerce Customer Behavior Analysis.
Business Goal	Use analytics to understand the business problem and support better decisions related to E-Commerce Customer Behavior Analysis.
Tools Used	Python, Power BI
Libraries	pandas, numpy, matplotlib, seaborn, scikit-learn
Techniques Applied	EDA, Data Visualization, KPI Analysis, RFM Segmentation, Predictive Analytics (where applicable)
Deliverables	Cleaned dataset, analysis notebook, interactive Power BI dashboard, insights and recommendations
Difficulty Level	Intermediate

1.2 Introduction

E-commerce platforms generate massive volumes of behavioral data at every stage of the customer journey — from the first site visit to browsing, adding items to cart, and finally completing (or abandoning) a purchase. This project analyzes a simulated e-commerce transaction dataset of 25,000 sessions from 8,442 unique customers to understand how customers behave, what drives revenue, and where the business is losing potential sales. The analysis combines Python-based exploratory data analysis (EDA) with an interactive Power BI dashboard to deliver a complete, end-to-end view of customer behavior.

1.3 Objectives

- Clean and prepare the raw e-commerce dataset for analysis.
- Perform exploratory data analysis (EDA) to understand customer, product, and revenue patterns.
- Calculate key performance indicators (KPIs) such as conversion rate, cart abandonment rate, and average order value.
- Segment customers using RFM (Recency, Frequency, Monetary) analysis to identify high-value customers.
- Build an interactive Power BI dashboard for business stakeholders to explore the data visually.
- Translate the analysis into actionable business insights and recommendations.

1.4 Dataset Description

The dataset (Ecommerce.csv) contains 25,000 rows and 29 columns, with no missing values and no duplicate records. It captures session-level detail including customer demographics, device and marketing channel, product and pricing information, on-site behavior (pages viewed, time on site), the purchase funnel (added to cart, purchased, cart abandoned), post-purchase feedback (rating, review), and time-based fields (visit date, month, weekday, season).

Category	Key Columns
Customer & Session	customer_id, session_id, device_type, user_type, location
Acquisition	marketing_channel, payment_method
Product & Pricing	product_id, product_category, unit_price, quantity, discount_percent, discount_amount, revenue
On-site Behavior	pages_viewed, time_on_site_sec, session_duration_bucket
Purchase Funnel	added_to_cart, purchased, cart_abandoned
Feedback	rating, review_text, review_helpful_votes
Time	visit_date, visit_day, visit_month, visit_weekday, visit_season

2. Methodology

The analysis followed a structured data analytics workflow, moving from raw data ingestion through cleaning, exploratory analysis, KPI computation, customer segmentation, and finally dashboard reporting.

2.1 Data Cleaning & Preparation (Python)

- Loaded the raw CSV (25,000 rows × 29 columns) into a pandas DataFrame.
- Checked for missing values across all 29 columns — none were found.
- Checked for duplicate rows — none were found, confirming data integrity.
- Converted the visit_date column from text to a proper datetime format for time-based analysis.
- Validated that revenue, quantity, and unit_price contained no negative or invalid values.

2.2 Analysis Workflow

- Exploratory Data Analysis (EDA) using pandas, matplotlib and seaborn to uncover customer, product and revenue trends.
- KPI computation — conversion rate, cart abandonment rate, add-to-cart rate and repeat-customer rate.
- RFM (Recency, Frequency, Monetary) segmentation to classify customers into value-based groups.
- Correlation analysis to identify which behavioral factors most strongly influence revenue and purchase decisions.
- Dashboard design in Power BI to present the findings in an interactive, business-friendly format.

3. Python Exploratory Data Analysis (EDA)

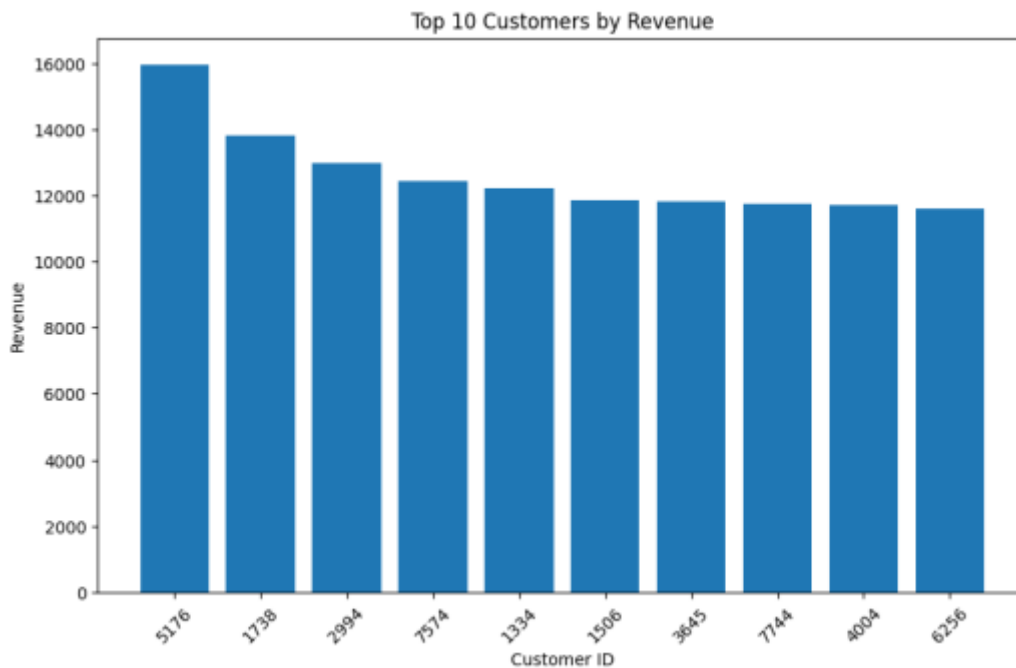
This section presents the key visualizations generated in Python (pandas, matplotlib, seaborn) during exploratory analysis, along with the interpretation of each chart.

3.1 Overall Business Snapshot

Metric	Value
Total Customers	8,442
Total Sessions	25,000
Total Revenue	₹10,116,169.06 (~10.12M)
Average Revenue per Session	₹404.65
Conversion Rate	22.46%
Add-to-Cart Rate	64.47%
Cart Abandonment Rate	42.00%
Repeat Customer Percentage	81.95% (6,918 repeat vs 1,524 one-time)

Roughly 82% of customers return for more than one session, indicating strong platform stickiness. However, only 22.46% of sessions convert into a purchase, and 42% of carts are abandoned — pointing to a clear gap between interest and completed transactions that the business can work to close.

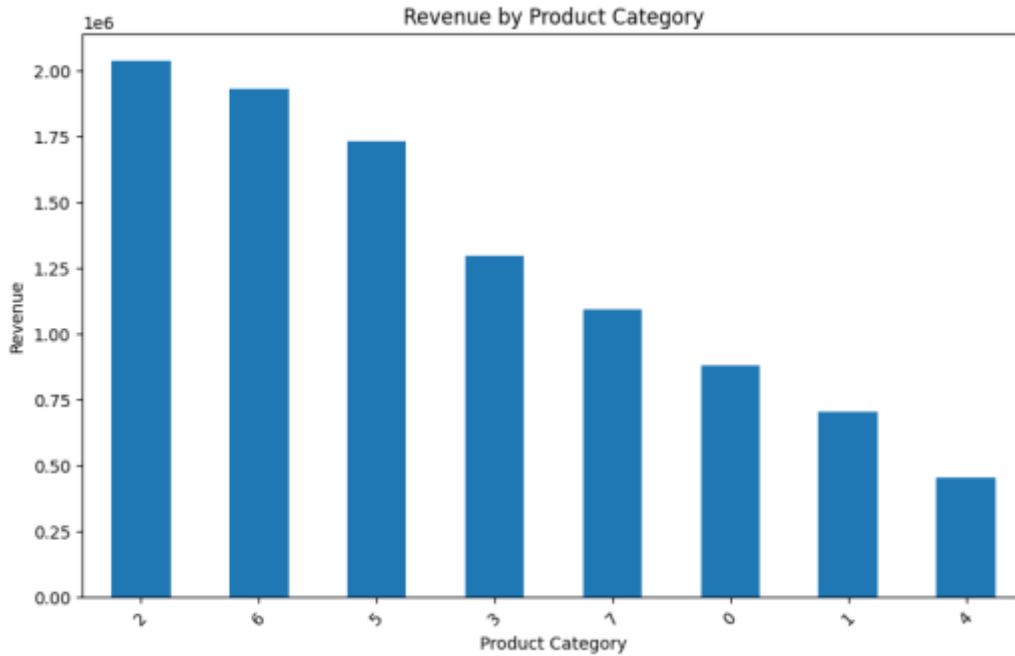
3.2 Top 10 Customers by Revenue



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A small group of customers drives disproportionately high revenue. The top customer (ID 5176) alone contributed close to ₹16,000, nearly 15% more than the tenth-ranked customer. This long-tail pattern confirms that a loyalty or VIP program targeting the top 1–2% of customers could meaningfully protect and grow revenue.

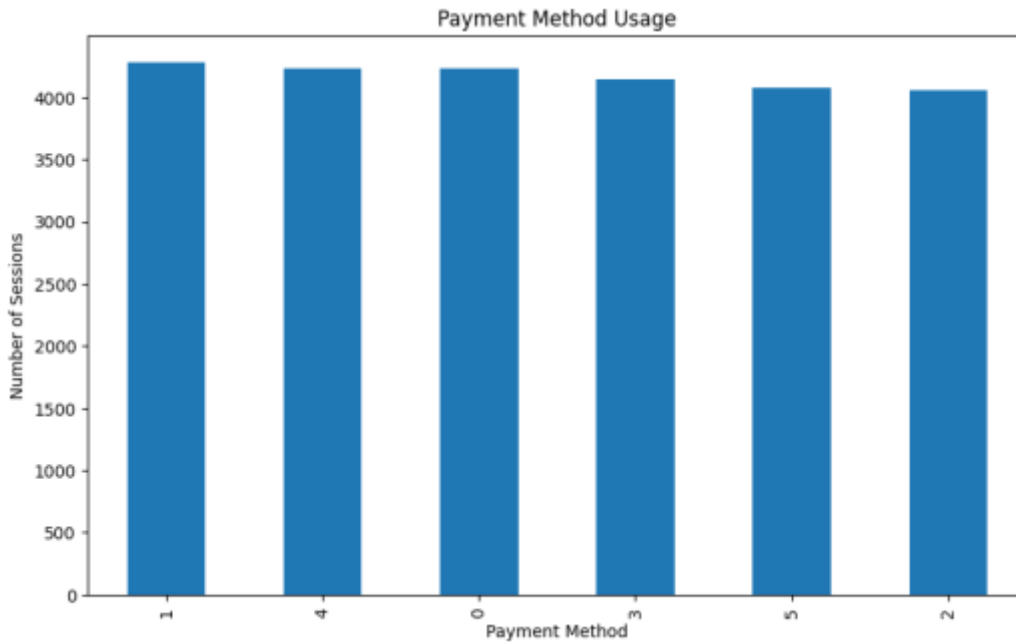
3.3 Revenue by Product Category



3.3 Revenue by Product Category

Category 2 generates the highest revenue (₹2.04M), followed by Category 6 (₹1.93M) and Category 5 (₹1.73M), while Category 4 trails all categories at just ₹452K. Interestingly, when compared against quantity sold (Category 2 also leads in units, but Category 4 sells a similar volume to top categories yet earns far less revenue), this suggests Category 4 products are priced significantly lower or discounted more heavily, and may need a pricing or bundling review.

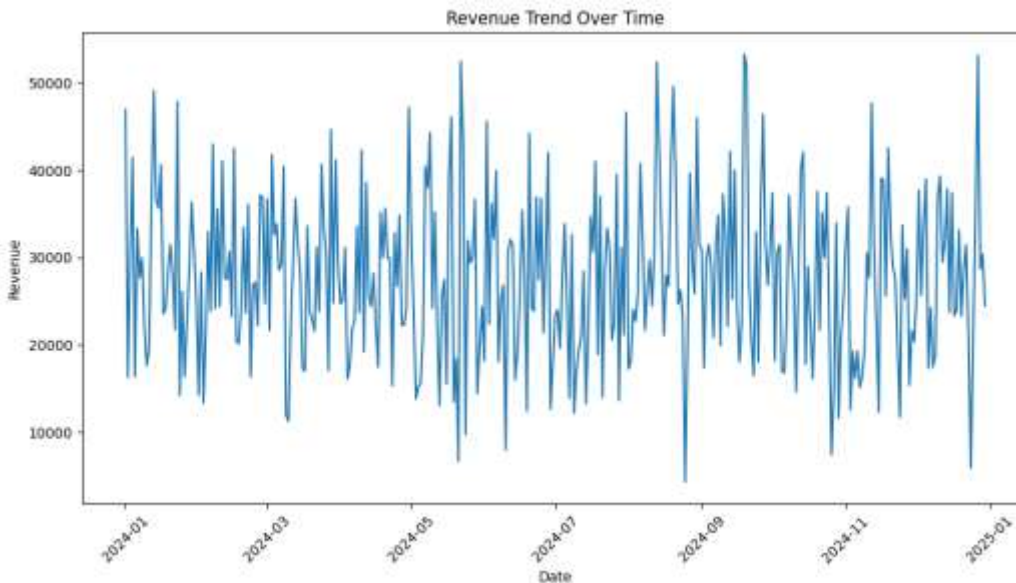
3.4 Payment Method Usage



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Usage is fairly evenly distributed across all six payment methods (ranging from 4,051 to 4,281 sessions each), with method 1 slightly ahead. No single payment method dominates, indicating customers are comfortable with a variety of payment options and the platform does not need to prioritize any one payment gateway over another.

3.5 Revenue Trend Over Time

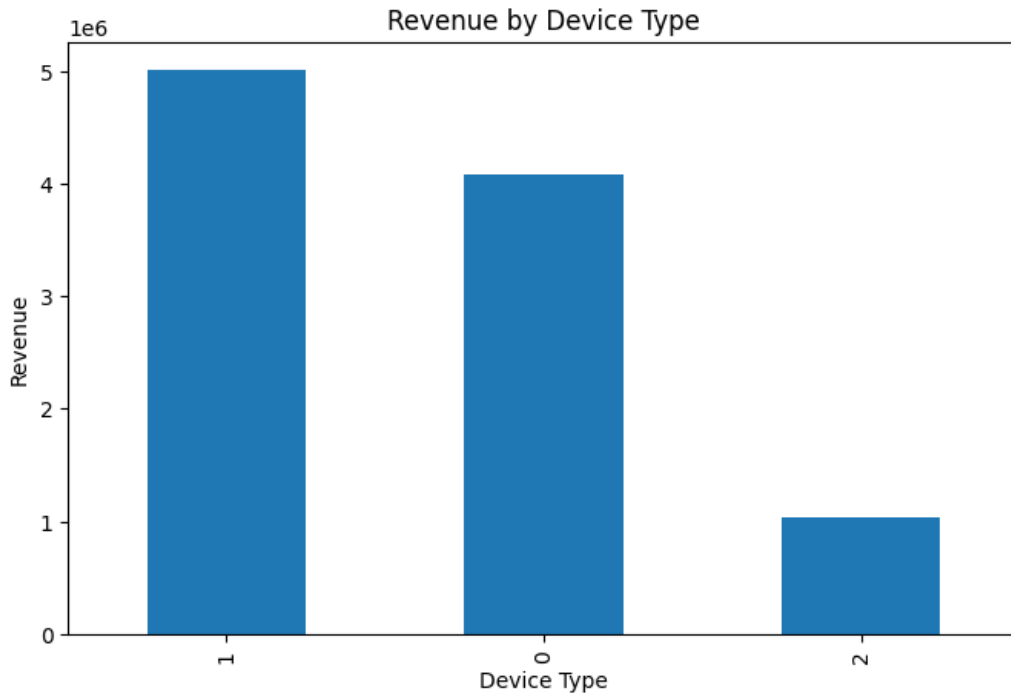


3.5 Revenue Trend Over Time

Daily revenue fluctuates between roughly ₹5,000 and ₹53,000 throughout the year with no strong long-term upward or downward trend, but with clear day-to-day volatility and periodic spikes (visible around May,

August–September, and late December). This volatility is likely tied to promotions, weekends, or seasonal campaigns and suggests revenue could be smoothed and grown with a more consistent promotional calendar.

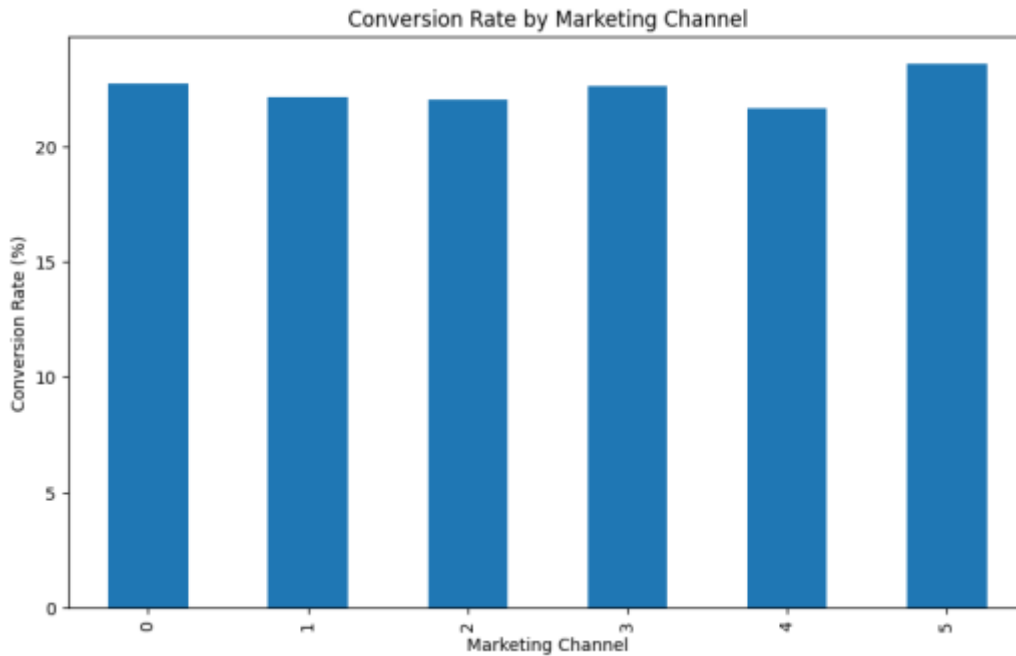
3.6 Revenue by Device Type



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Device type 1 generates the highest revenue (₹5.01M), followed by device type 0 (₹4.08M), while device type 2 contributes only ₹1.03M. This mirrors the Power BI donut chart (45.05% vs 40.21% vs 14.73% of customers), confirming that whichever platform device '1' represents (likely mobile or desktop) should be the primary focus for UX optimization and testing.

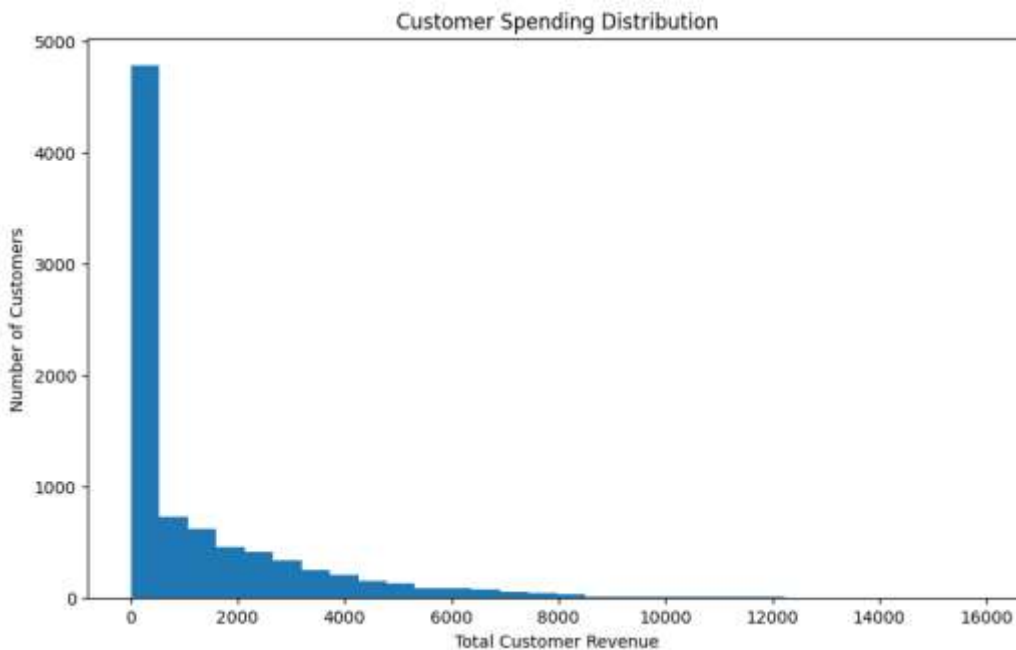
3.7 Conversion Rate by Marketing Channel



3.7 Conversion Rate by Marketing Channel

Conversion rates are close across all six marketing channels (21.66%–23.60%), with channel 5 converting best (23.60%, 1,006 purchases) and channel 4 the weakest (21.66%). While the gap is modest, channel 5 also produced the highest revenue (₹1.82M) among all channels, making it the strongest overall performer and a candidate for increased marketing spend.

3.8 Customer Spending Distribution



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The distribution is heavily right-skewed: the large majority of customers have generated under ₹2,000 in total revenue, while a small number of customers extend out to ₹12,000–16,000+. This classic 'long tail' pattern

reinforces the earlier finding from the Top-10-Customers chart — a small, high-value customer segment is disproportionately important to the business.

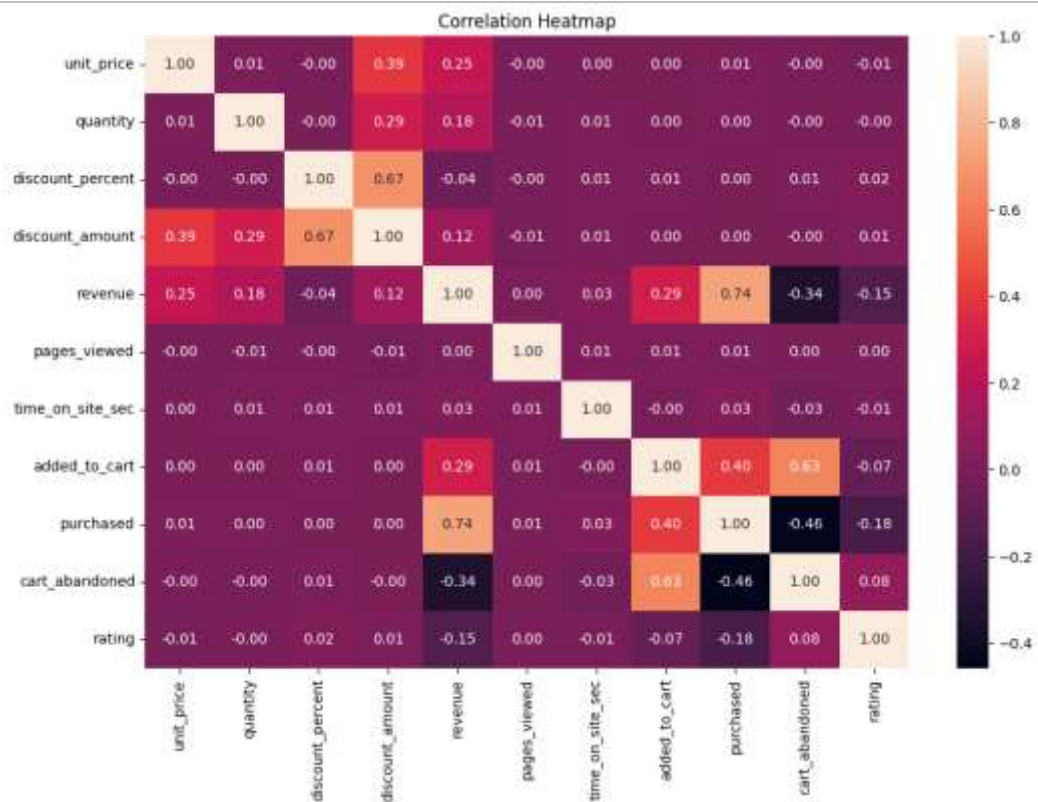
3.9 Purchase Frequency vs Customer Spending



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There is a positive relationship between number of sessions and total revenue up to around 6 sessions, after which returns diminish and even decline slightly for very high-frequency customers (9–11 sessions). This suggests that customer value is not purely driven by visit frequency; the most profitable customers tend to be moderately frequent, high-intent purchasers rather than customers who simply browse often without high spend.

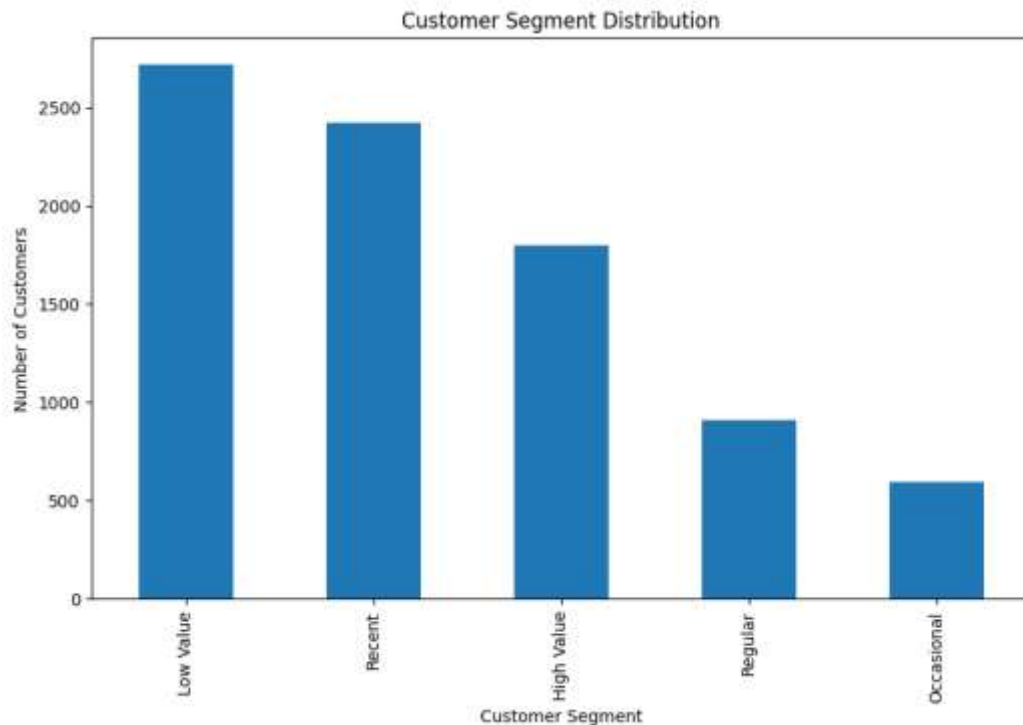
3.10 Correlation Heatmap



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The strongest positive correlation is between revenue and purchased (0.74), confirming purchases are the main revenue driver, as expected. `added_to_cart` correlates positively with both `purchased` (0.40) and `cart_abandoned` (0.63), while `purchased` and `cart_abandoned` are negatively correlated (-0.46), which is logical since a completed purchase and an abandoned cart are largely mutually exclusive outcomes of the same session. Pricing variables (`unit_price`, `discount_percent`, `discount_amount`) show weak correlation with revenue and purchase behavior, suggesting price and discounting alone are not the primary drivers of conversion — on-site engagement (adding to cart) matters more.

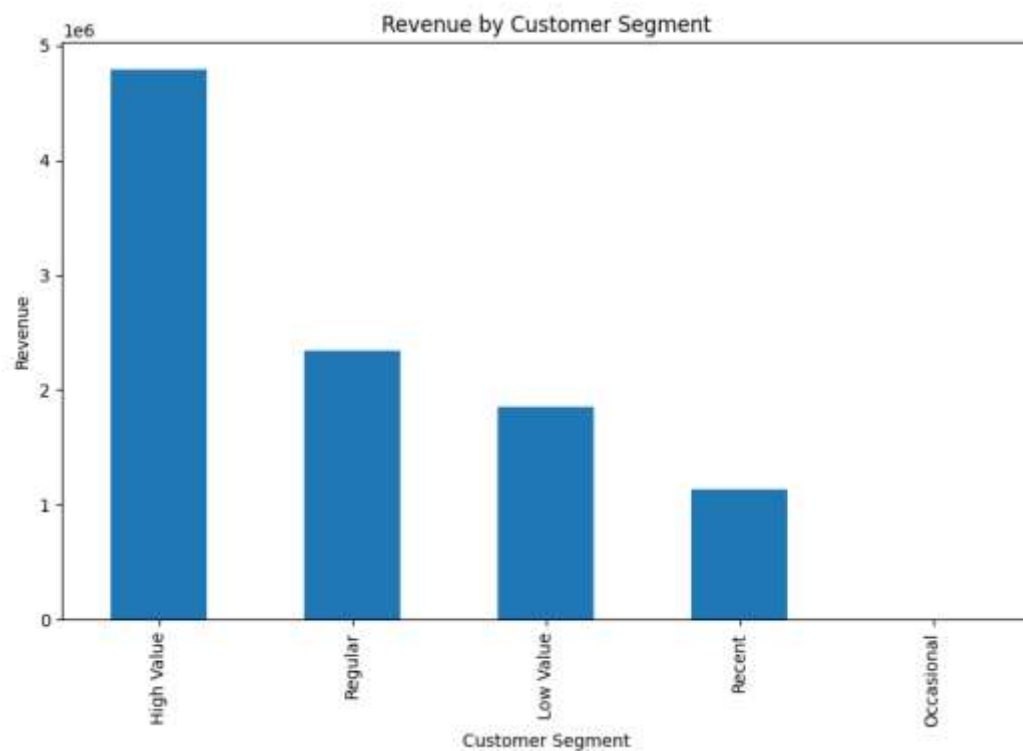
3.11 Customer Segment Distribution (RFM)



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Using RFM (Recency, Frequency, Monetary) scoring, customers were classified into five segments. 'Low Value' is the largest segment (2,719 customers), followed by 'Recent' (2,422), 'High Value' (1,799), 'Regular' (908) and 'Occasional' (594, customers with zero recorded purchase revenue).

3.12 Revenue by Customer Segment (RFM)



3.12 Revenue by Customer Segment (RFM)

Despite being smaller in count than the 'Low Value' and 'Recent' segments, the 'High Value' segment (1,799 customers, ~21% of the base) generates by far the most revenue at ₹4.79M — more than double the 'Regular' segment (₹2.34M) and far more than the much larger 'Low Value' segment (₹1.85M). This is a textbook 80/20 pattern and strongly supports prioritizing retention and loyalty investment in the High Value segment, while the zero-revenue 'Occasional' segment (594 customers) represents an at-risk / re-engagement opportunity.

4. Power BI Dashboard

An interactive Power BI dashboard titled 'E-Commerce Analytics' was built on top of the cleaned dataset (loaded as the Ecommerce_Data table, 25,000 rows) to give business stakeholders a live, filterable view of the same metrics explored in Python.

4.1 Dashboard Overview



Figure: E-Commerce Analytics — Power BI Dashboard (Page 1)

The dashboard opens with six KPI cards summarizing the entire dataset at a glance: Total Customers (8.442K), Total Sessions (25K), Total Revenue (10.12M), Total Purchases (6K), Total Quantity Sold (4, shown as an average), and Average Revenue (404.65). These figures match exactly with the Python-computed KPIs, validating that both tools are reading from the same underlying data and confirming the consistency of the analysis.

Customer by Device Type (Donut Chart)

The donut chart breaks down the 8.442K customers by device_type: type '1' accounts for 45.05% (6.75K), type '0' accounts for 40.21% (6.03K), and type '2' accounts for only 14.73% (2.21K). This distribution matches the Python revenue-by-device chart almost exactly, reinforcing that device type '1' is both the most common and

the highest-revenue-generating platform, and should be the priority for design, performance, and UX investment.

Sum of Marketing Channel by Customer ID

This visual plots marketing channel values across individual customer IDs. Because it is rendered at the individual customer level rather than aggregated by channel, it mainly shows the spread and density of channel assignment across the customer base rather than a clean comparison — the aggregated Python view (Section 3.7) is more directly actionable for channel-level decisions.

Sum of Revenue by Month

The monthly revenue line shows clear peaks and troughs across the year, ranging from roughly 800K to just under 950K per month, with the two highest peaks occurring mid-year and again in the September–October period. This seasonal fluctuation is consistent with the day-level volatility seen in the Python revenue trend chart and suggests recurring promotional or seasonal effects worth planning around.

Sum of Revenue by Visit Season

The horizontal bar chart shows revenue by encoded season value (0–3), with season '3' generating the highest revenue and season '0' the lowest, a difference of roughly 1M between the strongest and weakest season. This supports allocating a larger share of the marketing budget to the higher-revenue season(s).

4.2 DAX Measures Created

A set of DAX measures was created in the Ecommerce_Data table to power the dashboard's KPI cards and enable flexible aggregation:

Measure	DAX Formula	Purpose
Total Quantity	SUM(Ecommerce_Data[quantity])	Total units sold across all sessions
Total Customers	DISTINCTCOUNT(Ecommerce_Data[customer_id])	Unique customer count
Cart Abandoned	CALCULATE(COUNTROWS(Ecommerce_Data), Ecommerce_Data[cart_abandoned] = 1)	Count of sessions where the cart was abandoned
Total Revenue	SUM(Ecommerce_Data[revenue])	Total revenue generated

The screenshot shows a Tableau Desktop interface with a table calculation named 'Total Quantity' (SUM of quantity). The dimensions are: customer_id, session_id, date, device_type, user_type, marketing_channel, product_id, product_category, unit_price, quantity, discount_percent, and discount_amount. The table displays data for various sessions and products, with a total quantity of 1,000.00 for the 'Total Quantity' measure.

customer_id	session_id	date	device_type	user_type	marketing_channel	product_id	product_category	unit_price	quantity	discount_percent	discount_amount
8000	3011	21 June 2020	P	0	0	648	0	730.60	0	0	0
8127	643	11 April 2020	P	1	0	981	0	6204.75	1	0	0
9109	980	12 April 2020	P	1	0	46	0	1993.77	0	0	0
4802	894	19 November 2019	P	1	0	143	0	1138.79	0	0	0
5750	894	30 May 2020	P	0	0	229	0	441.81	0	0	0
3014	1228	20 January 2020	P	1	0	241	0	5475.14	0	0	0
9014	2034	10 August 2020	P	1	0	657	0	1562.21	1	0	0
1577	2128	16 April 2020	P	0	0	277	0	104.69	1	0	0
2800	4164	17 February 2020	P	1	0	649	0	790.73	0	0	0
9010	2751	18 June 2020	P	0	0	894	0	1805.14	0	0	0
8111	2898	19 March 2020	P	0	0	307	0	1147.14	0	0	0
9000	3032	29 November 2019	P	0	0	276	0	1344.73	0	0	0
3011	2621	31 January 2020	P	1	0	753	0	276.64	0	0	0
9014	2805	20 January 2020	P	1	0	193	0	1149.23	0	0	0
1510	4704	05 November 2020	P	0	0	64	0	1900.87	0	0	0
2800	4164	20 June 2020	P	1	0	628	0	121.3	0	0	0
8110	4903	22 February 2020	P	0	0	614	0	487.90	0	0	0
1777	1018	02 June 2020	P	1	0	379	0	487.91	0	0	0
9803	1863	21 Aug 2020	P	1	0	398	0	1000.90	0	0	0
4011	0236	02 February 2020	P	0	0	803	0	106.1	0	0	0
4200	0099	26 October 2020	P	1	0	20	0	362.0	0	0	0
9015	4400	19 March 2020	P	0	0	259	0	1607.82	1	0	0
8020	6779	19 January 2020	P	0	0	891	0	477.77	0	0	0
8902	9821	18 April 2020	P	0	0	578	0	807.62	0	0	0
8147	5044	11 February 2020	P	0	0	258	0	271.16	1	0	0
4375	0348	26 June 2020	P	0	0	101	0	147.42	1	0	0
5340	3276	23 April 2020	P	0	0	344	0	498.22	0	0	0
8891	2708	17 March 2020	P	0	0	52	0	1918.20	0	0	0
1016	4740	20 April 2020	P	1	0	376	0	477.60	0	0	0

Figure: Total Quantity measure (SUM of quantity)

The screenshot shows a Tableau Desktop interface with a table calculation named 'Total Customers' (DISTINCTCOUNT of customer_id). The dimensions are: customer_id, session_id, date, device_type, user_type, marketing_channel, product_id, product_category, unit_price, quantity, discount_percent, and discount_amount. The table displays data for various sessions and products, with a total of 1,000 distinct customers.

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Figure: Total Customers measure (DISTINCTCOUNT of customer_id)

Figure: Cart Abandoned measure (CALCULATE + COUNTROWS)

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Figure: Total Revenue measure — an initial version referencing an incorrectly named table ('Ecommerce') produced a 'Cannot find table' error, which was corrected by pointing the formula to the actual table name, Ecommerce_Data.

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Note: The Total Revenue screenshot above captures an intermediate debugging step where the measure referenced the table name 'Ecommerce' instead of the correct table name 'Ecommerce_Data', triggering Power BI's 'Cannot find table' validation error. This was corrected in the final model, and the working formula (SUM(Ecommerce_Data[revenue])) is what powers the 10.12M Total Revenue KPI card shown on the dashboard.

5. Key Insights & Recommendations

5.1 Key Insights

- Revenue is concentrated: a small set of high-value customers (the 'High Value' RFM segment, ~21% of customers) generates the majority of revenue (₹4.79M of ₹10.12M total, ~47%), following an 80/20-style pattern.
- Conversion has room to grow: only 22.46% of sessions result in a purchase, and 42% of carts are abandoned — nearly double the purchase rate — representing significant recoverable revenue.
- Device type matters: device type '1' drives both the largest customer share (45%) and the highest revenue (₹5.01M), making it the top platform priority.
- Category performance is uneven: Category 2 leads in both units sold and revenue, while Category 4 sells a comparable volume to top categories but earns the least revenue, pointing to a pricing or product-mix issue.
- Marketing channels perform similarly, but channel 5 stands out with both the highest conversion rate (23.6%) and highest revenue (₹1.82M), making it the most efficient channel.
- On-site engagement (adding to cart) is more strongly linked to revenue and purchase behavior than pricing or discounting, based on the correlation analysis.
- Customer loyalty is strong: 81.95% of customers return for more than one session, but a segment of 594 'Occasional' customers has generated zero revenue and represents a re-engagement opportunity.

5.2 Recommendations

- Launch a loyalty / VIP program targeting the High Value RFM segment to protect and grow the ~47% of revenue they already generate.
- Investigate and address cart abandonment (42%) through targeted cart-recovery emails, simplified checkout, or limited-time incentives.
- Prioritize UX, performance, and A/B testing on device type '1', since it accounts for the largest customer base and highest revenue.
- Review pricing and promotional strategy for Category 4 products, which underperform on revenue despite healthy sales volume.
- Increase marketing investment in channel 5, the best-performing channel on both conversion rate and revenue.
- Design a re-engagement campaign (e.g., personalized offers) for the 594 'Occasional' customers who have browsed but never converted.
- Align promotional and marketing calendars with the observed monthly/seasonal revenue peaks to smooth out revenue volatility.

6. Conclusion

This project applied a complete data analytics workflow — from data cleaning and exploratory analysis in Python to KPI computation, RFM customer segmentation, and interactive dashboard reporting in Power BI — to a 25,000-session e-commerce dataset. The analysis consistently showed that a relatively small group of highly engaged, high-value customers drives a disproportionate share of revenue, that a substantial share of purchase intent is lost to cart abandonment, and that device type and marketing channel choice measurably affect performance. The Python analysis and the Power BI dashboard produced matching KPIs (8,442 customers, 25,000 sessions, ₹10.12M revenue, ₹404.65 average revenue), validating the reliability of the pipeline end-to-

end. The resulting insights and recommendations provide a concrete, data-backed foundation for improving conversion, customer retention, and revenue growth on the platform.

Deliverables produced as part of this project: the cleaned dataset, the Python analysis notebook (Google Colab), the interactive Power BI dashboard, and this consolidated insights and recommendations report.